07. Stress-tensor bootstrap: a sector-specific gap bound in three dimensions

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Energy and momentum must be accounted for consistently in a quantum theory. The stress tensor is the mathematical object that tracks them. By examining how four stress-tensor insertions relate to one another, we can test whether a proposed theory is missing an excitation it ought to contain. The question here is how large a particular gap in the spectrum can be before the consistency rules contradict each other.

Key terms

- **Stress tensor T** — The field that describes the local flow and distribution of energy and momentum.
- **Conformal field theory** — A quantum theory with symmetry under angle-preserving transformations, including rescaling lengths.
- **Central-charge parameter C_T** — The normalization of the stress tensor's two-point correlation, used to characterize the theory in a fixed convention.
- **TTT data** — Coefficients describing a three-stress-tensor correlation; they contain information beyond C_T.
- **Scalar sector** — The set of exchanged operators with zero spin, chosen as the focus of the gap question.
- **Parity-even** — Unchanged in sign under the specified spatial-reflection transformation.
- **Spectral gap** — Here, the threshold in scaling dimension before the first nonidentity operator in the selected sector appears, not necessarily a material's energy gap.
- **Tensor structure** — One of the distinct ways the directional indices of spinning fields can enter a correlation function.
- **Positive semidefinite matrix** — A matrix whose quadratic form is nonnegative for every vector; it expresses the needed positivity for multiple coupled tensor structures.

Why this matters

The stress tensor is present in a broad class of local theories, so restrictions derived from it can have wide reach. Such restrictions help map which strongly interacting quantum theories are possible. They can also constrain proposed boundary descriptions of gravity, where the stress tensor has a special role.

What progress would mean

A verified gap bound would show that a theory with the declared inputs must contain a certain kind of excitation below a threshold. It would not determine the entire spectrum or prove that a gravity theory is unique.

Core question

At fixed stress-tensor two- and three-point data, can crossing exclude a proposed gap in the parity-even scalar sector of T×T in a unitary three-dimensional CFT?

Scope and assumptions

Fix d=3, parity invariance, a unique conserved stress tensor, a normalization for C_T, and the remaining independent TTT parameter. The initial observable is the first parity-even nonidentity scalar appearing with nonzero coupling in T×T. It is not a bound on every operator in the theory. State separate gaps for any other restricted spin/parity sectors.

Use the full conserved spinning correlator. A scalar block toy implementation is permitted for learning but is not the target stress-tensor bootstrap.

Mathematical target

Write the crossing equations in a nonredundant tensor basis as

$$V_{id} + V_T(C_T, t_4) + \sum_{\mathcal{O}} \lambda_{\mathcal{O}}^{\mathsf{T}} V_{\Delta,\ell,\pi} \lambda_{\mathcal{O}} = 0.$$

The OPE vectors account for multiple tensor structures. In d=3, a conserved symmetric-traceless spin-ℓ primary has $\Delta=\ell+1$, not $\Delta=3$ for all spins. For a proposed scalar gap g, seek α with $\alpha(V_{id}+V_T)>0$ and $\alpha(V_{\{\Delta,\ell,\pi\}})$ positive semidefinite on every allowed exchange. This yields a contradiction and therefore excludes g under the specified assumptions.

Required outputs and proof obligations

Export tensor/block conventions, conservation and Ward identities, fixed input data, derivative basis, spin treatment, and functional coefficients. Attach rigorous block-approximation and high-spin control. If only the finite approximant is checked, label the result accordingly.

Validation and rejection controls

Recover a selected published stress-tensor benchmark using exactly its conventions and assumptions. Check the allowed region rather than the excluded region. Verify matrix positivity, not just individual scalar components. Free scalar/fermion data provide normalization tests; numerical agreement alone is not a proof of the continuum bound.

Milestones and research extension

Baseline: reproduce one finite benchmark and publish the assumptions. Strong: turn one exclusion into a certified result with omitted-sector control. Research extension: improve a bound, add another sector, or obtain a conditional extremal spectrum. Saturation does not construct a CFT.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Dymarsky et al., The 3d Stress-Tensor Bootstrap — Conserved tensor crossing and sector-specific numerical gap bounds.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$V_{\mathrm{id}} + V_T(C_T, t_4) + \sum_{\mathcal{O}} \lambda_{\mathcal{O}}^{\mathsf{T}} V_{\Delta,\ell,\pi} \lambda_{\mathcal{O}} = 0.$$

$$\alpha(V_{\mathrm{id}} + V_T) > 0, \qquad \alpha(V_{\Delta,\ell,\pi}) \succeq 0 \quad \text{for every allowed exchange.}$$

These inequalities contradict crossing and exclude the proposed sector-specific gap. Tensor structures, conserved spinning blocks, Ward identities and continuous exchange coverage are required. In three dimensions a conserved symmetric-traceless spin-ℓ primary has $\Delta = \ell + 1$.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — incorrect current dimensions (main original 60–64).** Higher-spin conserved currents have Δ=ℓ+d−2. Assigning all of them Δ=d violates the representation-theory input.
2. **Blocking — scalar treatment of spinning crossing (49–75).** TTTT has multiple tensor structures and OPE matrices; a single squared coefficient and G(u,v)=G(v,u) do not specify its crossing system.

3. **Blocking — functional domain/sign (83–91; short duplicate 39–49).** Positivity must hold for allowed operators. The original asks for positivity in the excluded region and alternates lower and upper gap bounds. The rewrite gives an explicit contradiction convention.
4. **Major — ambiguous target and duplicate (both 07 files).** A gap to a conserved current, a scalar gap, and a single-trace holographic gap are different observables. Both originals now resolve to one canonical scalar-sector problem; gravity uniqueness is not inferred from it.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **At fixed stress-tensor two- and three-point data, can crossing exclude a proposed gap in the parity-even scalar sector of T×T in a unitary three-dimensional CFT?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: reproduce one finite benchmark and publish the assumptions. Strong: turn one exclusion into a certified result with omitted-sector control. Research extension: improve a bound, add another sector, or obtain a conditional extremal spectrum. Saturation does not construct a CFT.

Sources supporting the corrections

- Dymarsky et al., The 3d Stress-Tensor Bootstrap — Conserved tensor crossing and sector-specific numerical gap bounds.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented

trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.

- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.